

1. Which of the following best describes the physical quantity represented by the gradient of a velocity–time graph for an object undergoing motion along a straight line?
 - a. Total displacement covered by the object
 - b. Instantaneous rate of change of position
 - c. Net force acting on the object
 - d. Rate of change of velocity with respect to time
2. Which of the following modifications would most effectively minimize systematic and random errors, thereby reducing the deviation between experimental measurements and theoretical predictions?
 - a. Enhancing the resolution and calibration of measuring instruments
 - b. Intentionally increasing resistive forces along the track
 - c. Selectively omitting inconsistent data points without justification
 - d. Altering the accepted value of gravitational acceleration
3. In the context of analyzing motion on an inclined plane, the parameter h/L is most appropriately interpreted as:
 - a. The absolute vertical displacement of the system
 - b. The total path length traveled by the object
 - c. The instantaneous velocity ratio along the track
 - d. A dimensionless quantity representing the sine of the incline angle
4. Assuming negligible friction, how does an increase in the vertical elevation h of a fixed-length inclined plane affect the acceleration of a cart, based on the relation $a = gh/L$?
 - a. It increases the acceleration proportionally to the incline's sine component
 - b. It reduces the acceleration to increased travel distance
 - c. It produces no change since gravitational acceleration is constant
 - d. It reduces the velocity of the cart to zero at all points
5. In the experimental setup for investigating motion on an inclined plane, the vertical elevation of the track is adjusted to approximately which value to ensure measurable acceleration while minimizing excessive speed?
 - a. 0.05 m
 - b. 0.20 m
 - c. 0.10 m
 - d. 0.50 m
6. When determining the vertical height of the inclined track for accurate calculation of h/L , the measurement must be taken:
 - a. Between the laboratory floor and the highest support point
 - b. Along the upper surface where the cart travels
 - c. From the instantaneous position of the cart during motion
 - d. Perpendicular to the underside of the track to ensure correct geometric height
7. Based on the theoretical model for motion along a frictionless inclined plane, which expression correctly relates the acceleration of the object to gravitational acceleration and the geometry of the incline?
 - a. $a = g/L$
 - b. $a = g(h/L)$
 - c. $a = gh/L$
 - d. $a = g/hL$

8. In the context of Earth-based laboratory experiments, which value is conventionally adopted for the magnitude of gravitational acceleration when calculating theoretical results?

- a. 8.91 m/s^2
- b. 10.0 m/s^2
- c. 9.81 m/s^2
- d. 9.18 m/s^2

9. In an idealized inclined plane experiment with negligible friction, what fundamental relationship is being examined when additional masses are incrementally added to the cart while measuring its acceleration?

- a. The contribution of air resistance as mass increases
- b. The dependence of thermal energy transfer on motion
- c. Variations in the net applied force independent of gravitational influence
- d. The independence of gravitational acceleration from the mass of the object

10. Which procedure provides the most reliable determination of the cart's acceleration from experimental data collected during motion along the incline?

- a. Directly recording elapsed time intervals without graphical analysis
- b. Interpreting graphical color variations in the plotted data
- c. Computing the gradient of the best-fit line on a velocity–time graph
- d. Estimating values by counting discrete measurement points

Answer Key:

- 1. d
- 2. a
- 3. d
- 4. a
- 5. b
- 6. d
- 7. b
- 8. c
- 9. d
- 10. c